Chapter 13

Quantifying Grain Digestibility of Starch Fractions in Milled Rice

Crisline Mae Alhambra, Sushil Dhital, Nese Sreenivasulu, and Vito M. Butardo Jr.

Abstract

Rice is one of the staple foods which serves as the major source of carbohydrate in the human diet. A typical milled rice grain is mainly composed of starch of up to 80–90%, with an average of 6–8% proteins and some trace amounts of dietary fiber. Although cooked white rice can elicit variable glycemic response, a portion of rice starch may evade digestion in the human small intestine. The digested portion of rice can be estimated and characterized in vitro based on starch digestion extent and rate (kinetics). The indigestible portion of starch can also be quantified. This chapter will present micro-scale methods to quantify rice starch digestion rate and extent based on the sugar fractions released after treating the samples with digestive enzymes.

Key words Rice, Grain, Digestibility, Resistant Starch, Nutrition

1 Introduction

Rice is one of the major food crops in the world, feeding almost half of the world's population. The major component of rice grain is carbohydrates in the form of starch and some free sugars [1, 2]. Up to 70% of the human energy intake comes from carbohydrates. Starch can be classified into three fractions based on extent of hydrolysis after treatment with digestive enzymes: (1) Rapidly Digestible Starch (RDS), (2) Slowly Digestible Starch (SDS), and (3) Resistant Starch (RS). RDS and SDS are the fractions of starch which can be digested in the small intestine and subsequently absorbed by the body within 20 min and between 20 and 120 min, respectively. RDS represents rapidly released glucose after consumption of starchy food, while SDS has been associated with reduced sensation of hunger after ingestion due to the slowing down of carbohydrate hydrolysis rate [3]. The undigested fraction of starch is termed RS, which is resistant to small intestinal digestion but mostly fermentable in the colon by the gut microflora

Nese Sreenivasulu (ed.), *Rice Grain Quality: Methods and Protocols*, Methods in Molecular Biology, vol. 1892,
https://doi.org/10.1007/978-1-4939-8914-0_13, © Springer Science+Business Media, LLC, part of Springer Nature 2019

[4, 5]. Since RS does not contribute to raise in blood glucose level, it is known to help in the prevention of hyperglycemia, hyperinsulinemia, diabetes, and obesity [6–8].

Both underweight and obesity are major risk factors for the global burden of disease [9]. In both cases, there is a need to investigate the digestibility of foods, mainly the starch portion in common foods such as rice. In this connection, the identification of different fractions of starch in rice is important in determining its effect on human body upon consumption considering the prevalence of diet-induced nutritional concerns which ranges from caloric hunger on one end, to diabetes, overweight, and obesity on the other end of the spectrum.

This method focuses on identifying different starch fractions in polished rice grains to quantify and characterize rice grain digestibility. Microscale methods based on Megazyme kits are presented to determine the amount of total resistant starch (RS) as well as total digestible carbohydrates (DC) in one assay as well as total starch (TS) and total sugars (TSU) in a separate assay. These methods are optimum for large-scale screening of breeding and diversity populations of rice because they utilize a micro method wherein a tenfold decrease in the amount of sample and reagents are used to facilitate assays using microfuge tubes. Screening for digestibility properties in rice breeding and diversity panels are prohibitively expensive due to cost, time, and manpower involved. The methods outlined here greatly streamlined the process. Once superior germplasm is identified, the values obtained from the screening methods can be verified using the conventional AOAC methods for total starch (996-11) and resistant starch (2002-02). In addition, the cooked grain amylolysis method to determine the kinetics of starch digestibility simulating gastrointestinal tract digestion is presented in the last section as proxy measure for glycemic index (GI) measurements. Such an in vitro method helps to characterize diversity rice lines to estimate the values of cooked grain amylolysis. The presented cooked grain amylolysis method here can be used as a pre-screening method prior to actual clinical studies of GI estimation.

2 Materials

2.1 Kits

1. Resistant Starch Assay Kit (Megazyme K-RSTAR 08/11), 1000 assay per kit.
2. Total Starch Assay Kit (Megazyme K-TSTA 07/11), 1000 assay per kit.
3. D-Glucose Assay Kit (Megazyme K-GLUC 07/11), 660 assays per kit.

2.2 Reagents

1. 100 mM sodium maleate buffer pH 6.0 with 5 mM $CaCl_2 \cdot 2H_2O$ (500 mL)
2. 99% Ethanol (500 mL).
3. 50% ethanol (1 L).
4. 2 M KOH (250 mL).
5. 1.2 M sodium acetate buffer pH 3.8 (1 L).
6. 100 mM sodium acetate buffer pH 4.5 (5 L).
7. 80% ethanol (2 L).
8. 50 mM maleic acid buffer pH 5.5 (500 mL).
9. 0.01 M HCl (500 mL).
10. 0.02 M NaOH (500 mL).
11. 200 mM sodium acetate buffer pH 6 with 4 mM $CaCl_2$ and 0.49 mM $MgCl_2$ (1 L).
12. GOPOD reagent (1 L).

2.3 Instruments

1. Analytical balance (10 mg to 220 g capacity ±0.1 mg).
2. Refrigerated Microcentrifuge (4 °C, 24-microfuge tubes capacity).
3. Recirculating water bath (37 °C ± 0.03 °C temperature stability).
4. UV-Vis spectrophotometer or ELISA plate reader (510 nm).
5. Magnetic stirring plate (60–134 × *g*).
6. Speed vacuum dryer (36-microfuge holder).
7. Dehuller (100 g per minute).
8. Miller (8–10 g capacity).
9. Ball Grinder (grinding jar with 50 g capacity).
10. Sieve (30 and 40 mesh).

3 Methods

3.1 Preparation of Rice Samples

Dehull and mill the rice paddy using standard methods to obtain milled rice. Sort the milled rice and discard the broken and discolored grains. Grind polished grain samples (5 g) using a ball grinder set in 25 rpm per second for 1 min. In IRRI, we routinely use Retsch MM400 with 50 mL stainless steel grinder (01.462.0216) with a 25 mm diameter grinding ball (05.368.0105). Use this sample for resistant starch and total starch analysis. For cooked milled rice flour amylolysis, grind milled rice grains for 5 s using a ball grinder set in 20 rpm. Pass the ground sample between 30 and 40 mesh to collect particle size of 400 to 600 μm.

3.2 Total Resistant Starch (RS)

In this step, the total resistant starch (RS) is measured using the RS assay procedure from Megazyme (K-RSTAR 08/11) with modifications. This micro method had been extensively validated in our laboratory to accurately quantify RS comparable with the values obtained using the standard method. The supernatant is also collected to measure total digestible carbohydrates (DC) which is described in Subheading 3.3.

1. Weigh 10 ± 0.1 mg of rice flour directly in a 2 mL microfuge tube.
2. Add 400 μL of enzyme solution (10 mg/mL pancreatin with 3 U/mL amyloglucosidase (AMG) from the Megazyme kit dissolved in 100 mM sodium maleate buffer pH 6.0 with 5 mM $CaCl_2 \cdot 2H_2O$).
3. Incubate horizontally in a shaking water bath (100 strokes/min) at 37 °C for exactly 16 h. We commonly use microfuge rack to hold all the microfuge tubes in place using rubber bands.
4. Remove the tubes and dry each with a paper 12 min before the end of 16 h incubation.
5. Stop the reaction by the addition of 400 μL ethanol (99%).
6. Mix vigorously using a vortex mixer.
7. Centrifuge the tubes at 18,231 × *g* for 30 min.
8. After centrifugation, carefully transfer (*see* **Note 1**) the supernatant using a pipettor and collect in a 15 mL conical tube. Set aside for total digestible carbohydrates (DC) assay (Subheading 3.3).
9. Resuspend the pellets to 200 μL of 50% ethanol and vigorously stir using a vortex mixer.
10. Add a further 600 μL of 50% ethanol and vortex mix.
11. Centrifuge at 18,231 × *g* for 30 min.
12. Carefully remove the supernatant using a pipettor and repeat the suspension and centrifugation of the pellet (**steps 9–11**). Collect and pool all the supernatant obtained in the previous steps using the conical tube in **step 8**.
13. Prepare an ice bath over a stirring plate.
14. Place a stir bar in the tubes containing the pellet and add 200 μL of 2 M KOH while stirring in an ice bath. Stir for approximately 20 min or until complete solubilization of the pellet.
15. Add 800 μL of 1.2 M sodium acetate buffer pH 3.8 while stirring.
16. Immediately add 10 μL of 3300 U/mL of AMG and mix.

17. Place the tubes in water bath at 50 °C for 30 min with intermittent mixing (*see* **Note 2**).
18. After incubation, remove the stir bar and centrifuge at 18,231 × *g* for 10 min and get 100 μL of the supernatant (*see* **Note 3**).
19. Prepare a blank (100 μL of 100 mM sodium acetate buffer pH 4.5) and quadruplicate standards (100 μL of 1 mg/mL glucose, prepared and read four times).
20. Add 3 mL of GOPOD reagent supplied from the Megazyme kit to the 100 μL supernatant, blank and standards. Incubate at 50 °C for 20 min.
21. Read absorbance of the sample and standards against blank at 510 nm.
22. Calculate % RS (dry basis) by using the formula:

$$\%\text{RS} = \Delta A \times F \times \frac{1.03}{0.1} \times \frac{1}{1000} \times \frac{100}{W} \times \frac{162}{180}$$

where ΔA = absorbance of the sample against blank

$$F = \frac{1}{\text{average absorbance of glucose standard}}$$

$\frac{1.03}{0.1}$ = volume correction (0.1 mL aliquot was taken from 1.03 mL final volume)

$\frac{1}{1000}$ = conversion of μg to mg

$\frac{100}{W}$ = factor to present RS as % of sample weight

W = dry weigh of the sample = $\left(\text{flour weight} \times \frac{(100 \quad \text{moisture content})}{100}\right)$

$\frac{162}{180}$ = adjustment from free D-glucose to anhydrous D-glucose (as occurs in starch).

3.3 Total Digestible Carbohydrates (DC)

In this step, total digestible carbohydrates (DC) refer to total digestible starch and glucose present in the flour sample. The supernatant collected from Subheading 3.2 are used as starting materials.

1. Dilute the collected supernatant from RS assay (from Subheading 3.2, **step 12**) to 10 mL with 100 mM sodium acetate buffer pH 4.5.

2. Get 100 μL aliquot, and add 10 μL of AMG (300 U/mL in 100 mM sodium maleate buffer pH 6.0 with 5 mM $CaCl_2 \cdot 2H_2O$).
3. Incubate for 20 min at 50 °C.
4. Prepare a blank (110 μL of 100 mM sodium acetate buffer pH 4.5) and quadruplicate standards (110 μL of 1 mg/mL glucose, prepared and read four times).
5. Add 3 mL of GOPOD reagent the sample (**step 2**), blank and standards. Incubate at 50 °C for 20 min.
6. Read absorbance of the sample and standards against blank at 510 nm.
7. Calculate % DC (dry basis) by using the formula:

$$\%\ \mathrm{DC} = \Delta A \times F \times \frac{10}{0.1} \times \frac{1}{1000} \times \frac{100}{W} \times \frac{162}{180}$$

where

ΔA = absorbance of the sample against blank

$$F = \frac{100}{\text{average absorbance of glucose standard}}$$

$\frac{10}{0.1}$ = volume correction (0.1 mL aliquot was taken from 10 mL final volume)

$\frac{1}{1000}$ = conversion of μg to mg

$\frac{100}{W}$ = factor to present DC as % of sample weight

W = dry weigh of the sample = $\left(\text{flour weight} \times \frac{(100 - \text{moisture content})}{100}\right)$

$\frac{162}{180}$ = adjustment from free D-glucose to anhydrous D-glucose (as occurs in starch).

3.4 Total Starch (TS)

In this section, TS is measured using the Total Starch assay procedure (Megazyme K-TSTA 07/11) with modifications. This method had been extensively validated in our lab to accurately reflect the values obtained using the standard method. Total sugars (TSu) are assayed in Subheading 3.5 using the supernatant obtained in this section.

1. Weigh 10 ± 0.1 mg of starch directly in a 2 mL microfuge tube.
2. Add 500 μL of 80% ethanol and incubate at 85 °C for 5 min.
3. Stir using a vortex mixer after incubation and add 500 μL of 80% ethanol.
4. Mix and centrifuge at 18,231 × *g* for 20 min.
5. Aspirate (*see* **Note 4**) the supernatant and resuspend the pellet by adding 1 mL of 80% ethanol.
6. Mix and centrifuge at 18,231 × *g* for 20 min. Collect all the supernatant obtained for total free sugar quantification in a 2 mL microfuge tube (Subheading 3.5).
7. Prepare an ice bath over a stirring plate.
8. Place a stir bar to the obtained pellet and add 200 μL of 2 M KOH while stirring (700 rpm) in an ice bath.
9. Stir for approximately 20 min or until the complete solubilization of the pellet.
10. Add 800 μL of 1.2 M sodium acetate buffer pH 3.8 while stirring.
11. Immediately add 10 μL of thermo-stable α-amylase (3000 U/mL) and 10 μL of AMG (3300 U/mL).
12. Mix and incubate the tubes at 50 °C for 30 min (*see* **Note 5**).
13. Dilute the liquid contents with water to 10 mL (*see* **Note 6**).
14. Get 1 mL of aliquot, place in a 2 mL microfuge tube.
15. Centrifuge the aliquot at 18,231 × *g* for 10 min.
16. Get 100 μL of the supernatant, place in a separate tube.
17. Prepare a blank (100 μL H_2O) and quadruplicate standards (100 μL of 1 mg/mL glucose, prepared and read four times).
18. Add 3 mL GOPOD to the sample (**step 16**), blank and standards).
19. Incubate at 50 °C for 20 min.
20. Read absorbance of the sample and standards against blank at 510 nm.
21. Calculate % TS (dry basis) by using the formula:

$$\%\ \mathrm{TS} = \Delta A \times F \times \frac{10}{0.1} \times \frac{1}{1000} \times \frac{100}{W} \times \frac{162}{180}$$

where

ΔA = absorbance of the sample against blank

$$F = \frac{100}{\text{average absorbance of glucose standard}}$$

$\frac{1}{1000}$ = conversion of μg to mg

$$\frac{100}{W} = \text{factor to present TS as \% of sample weight}$$

W = dry weigh of the sample =

$$\left(\text{flour weight} \times \frac{(100 - \text{moisture content})}{100}\right)$$

$$\frac{10}{0.1} = \text{volume correction (0.1 mL was taken from 10 mL)}$$

$$\frac{162}{180} = \text{adjustment from free D-glucose to anhydrous D-glucose}$$

(as occurs in starch).

3.5 Total Free Sugar (TSu)

1. Dry the obtained supernatant from TS assay (Subheading 3.4) using a speed vacuum concentrator for approximately 4 h or until completely dry.
2. Reconstitute using 100 μL of 50 mM maleic acid buffer pH 5.5.
3. Add 50 μL of invertase (150 U/mL in 50 mM maleic acid buffer pH 5.5) and incubate for 30 min at 37 °C.
4. Boil in water bath for 4 min and cool immediately in ice bath.
5. Centrifuge at 13,394 × *g* for 10 min, get 50 μL of the supernatant and place in a 2 mL microfuge tube.
6. Prepare a blank (50 μL of 50 mM maleic acid buffer pH 5.5) and quadruplicate standards (50 μL of 1 mg/mL glucose, prepared and read four times).
7. Add 1.5 mL of GOPOD reagent to the sample (**step 5**), blank and standards.
8. Incubate at 50 °C for 20 min.
9. Read absorbance of the sample and standards against blank at 510 nm.
10. Calculate mg of glucose by using the formula:

$$\text{mg of glucose} = \frac{\Delta A}{F} \times \text{volume}$$

where

A = absorbance of the sample against blank

F = average absorbance of glucose standard

V = 0.150 mL

Calculate % sugar (glucose and sucrose) using the formula

$$\text{\% sugar} = \frac{\text{mg glucose}}{\text{wt of flour}} \times 100.$$

3.6 Cooked Rice Amylolysis

In this section, the starch digestion rate of cooked rice is determined by the use of enzymes and conditions that mimics the biochemical conditions of the human gastrointestinal digestion. Rice samples used have particle sizes between 400 and 600 μm to minimize the deviation arising from grain fragments, particle size and to mimic the size of chewed rice during mastication. Methods and calculations were developed based on Dhital et al. [10] and Butardo et al. [11].

1. Weigh 500 ± 0.1 mg milled rice (400–600 μm) in a 50 mL glass tube.
2. Add water (*see* **Note 7**) and cover with foil to minimize loss of water.
3. Let it stand for 30 min and cook in a boiling water bath for 30 min.
4. Immediately place vertically at 37 °C water bath over a stirring plate. If not used immediately, store at 50 °C waterbath (*see* **Note 8**). Ensure amylolysis will be done within the day.
5. Add 2 mL of water and dislodge the clumped cooked rice using a metal spatula. Add further 4 mL water to wash the spatula.
6. Add a stir bar (1/2 × 5/16 in) and start stirring at 350 rpm.
7. Immediately start the digestion process (*see* **Note 8**) by adding 5 mL of pepsin (*see* **Note 9**) (1 mg/mL in 0.01 M HCl) and incubate for exactly 30 min.
8. Add 5 mL of 2 M NaOH and incubate for exactly 1 min.
9. Add 20 mL of 200 mM sodium acetate buffer pH 6.0 with 4 mM $CaCl_2$ and 0.49 mM $MgCl_2$ and incubate for exactly 10 min. Get duplicate 200 μL aliquot 2 min before the end of the incubation.
10. Add 5 mL of pancreatin/AMG mixture (2 mg/mL in pancreatin and 28 U/mL AMG in 200 mM sodium acetate buffer pH 6.0 with 4 mM $CaCl_2$ and 0.49 mM $MgCl_2$) and get duplicate 200 μL aliquot at desired sampling points (5, 10, 20, 30, 45, 60, 90, 120, 180 min) and place in a microfuge tube.
11. Immediately place the microfuge tube containing the aliquot in an ice bath or liquid nitrogen to stop the reaction. Stopping the reaction can also be done using ethanol but slight modification in calculation has to be employed.
12. Centrifuge the aliquot at 15,720 × *g* at 4 °C for 10 min.
13. Aspirate 50 μL of the supernatant (*see* **Note 10**) and place in a 2 mL microfuge tube (*see* **Note 11**).
14. Prepare a blank (50 μL of 200 mM sodium acetate buffer pH 6.0 with 4 mM $CaCl_2$ and 0.49 mM $MgCl_2$) and qua-

druplicate standards (55 μL of 1 mg/mL glucose, prepared and read four times).

15. Add 5 μL of AMG (300 U/mL in 200 mM sodium acetate buffer pH 6.0 with 4 mM $CaCl_2$ and 0.49 mM $MgCl_2$) to both blank (**step 14**) and samples (**step 13**).
16. Incubate the samples, blank and standards at 50 °C for 20 min.
17. Add 3 mL GOPOD reagent and incubate at 50 °C for 20 min.
18. Read absorbance of the sample and standards against blank at 510 nm.
19. Calculate % starch hydrolysis (Sh) per sampling time by using the formula:

$$\% \text{ Sh} = \left\{ \frac{\left[\left(\frac{A\,\text{Sx}}{A\,\text{std}}\right) \times \text{d}f\right] \times V \times \frac{162}{180}}{\text{wt Sx} \times \frac{\%\ \text{TS}}{100}} \right\} \times 100$$

where:
AS_x = absorbance of the sample at time t
Astd = average absorbance of standard
df = dilution factor at time t
V = volume at time t (in mL)

$\frac{162}{180}$ = adjustment from free D-glucose to anhydrous D-glucose (as occurs in starch)
wt S_x = weight of sample (mg)
% TS = % total starch (dry basis).

20. Plot the % Sh vs time to get digestion curve.
21. Get the k-value by plotting $\ln[(\%\ \text{Sh}_{x+1} - \%\ \text{Sh}_x)/(t_{x+1} - t_x)]$, $\ln[(\%\ \text{Sh}_{x+2} - \%\ \text{Sh}_{x+1})/(t_{x+2} - t_{x+1})]$…etc. vs $(t_{x+1} - t_x)/2$, $(t_{x+2} - t_{x+1})/2$…etc.

    where:

    $x = 0$ min

    $x + 1 = 5$ min

    $x + 2 = 10$ min

    t = time (in the lab, we usually sample at 5, 10, 15, 20, 30, 45, 60, 90, 120, 180 min).
22. Get the equation of the line of the graph in **step 19**. The absolute value of the slope of the line (m) is the k value or digestion rate and is expressed as min^{-1}.

3.7 Integrating and Interpreting Results

RS and DC assay results would show the proportion of the starch in rice which is resistant to digestion and digestible, respectively. TS and TSu would show the total starch and soluble sugar (glucose and sucrose) contents of rice, respectively. Starch hydrolyzed within certain time points can be identified by using in vitro digestion using the cooked grain amylolysis method. In addition, the k value would determine the rate of digestion. Higher *k* value would indicate faster rate of digestion.

4 Notes

1. Use a 1 mL pipet to aspirate.
2. Mix during the first 15 min of incubation.
3. If RS is >10%, dilute the contents to 10 mL before centrifugation.
4. Similar to RS assay, use 1 mL pipet to aspirate.
5. Similar to RS assay, mix during the first 15 min of incubation.
6. Quantitatively transfer the contents by washing the stir bar with water.
7. Add water depending on amylose content (AC). The water:rice ratio is 1.8:1, 1.6:1 and 1.4:1 for samples with AC of 25+%, 20–24%, and 0–19%, respectively. Alternatively, one can just cook using excess water.
8. Do not allow the cooked rice to cool. Ideally, start the assay within 30 min to avoid starch retrogradation.
9. All enzyme solution to be used should be freshly prepared.
10. Store the supernatant at −20 °C if not used within the day.
11. Dilute the supernatant using 20 mM sodium maleate buffer pH 6.0 and dilution should be higher with increasing time. Get only 50 μL of the diluted supernatant.

References

1. Butardo VM Jr, Sreenivasulu N (2016) Tailoring grain storage reserves for a healthier rice diet and its comparative status with other cereals. Int Rev Cell Mol Biol 323:31–70
2. FAO (1998) Dietary carbohydrate composition. Food and Agriculture Organization of the United Nations, Rome
3. Sparti A, Milon H, Di Vetta V, Schneiter P, Tappy L, Jequier E, Schutz Y (2000) Effect of diets high or low in unavailable and slowly digestible carbohydrates on the pattern of 24-h substrate oxidation and feelings of hunger in humans. Am J Clinical Nut 72(6):1461–1468
4. Englyst HN, Cummings JH (1987) Digestion of polysaccharides of potato in the small-intestine of man. Am J Clin Nutr 45(2):423–431
5. Englyst HN, Macfarlane GT (1986) Breakdown of resistant and readily digestible starch by human gut bacteria. J Sci Food Agric 37(7):699–706
6. Ranganathan S, Champ M, Pechard C, Blanchard P, Nguyen M, Colonna P, Krempf

M (1994) Comparative-study of the acute effects of resistant starch and dietary-fibers on metabolic indexes in men. Am J Clin Nutr 59(4):879–883

7. Topping DL, Clifton PM (2001) Short-chain fatty acids and human colonic function: roles of resistant starch and nonstarch polysaccharides. Physiol Rev 81(3):1031–1064
8. Behall KM, Scholfield DJ, Hallfrisch JG, Liljeberg-Elmstahl HGM (2006) Consumption of both resistant starch and beta-glucan improves postprandial plasma glucose and insulin in women. Diabetes Care 29(5):976–981
9. World Health Organization (2002) The World Health Report: Reducing risks, promoting healthy life
10. Dhital S, Dabit L, Zhang B, Flanagan B, Shrestha AK (2015) In vitro digestibility and physicochemical properties of milled rice. Food Chem 172:757–765
11. Butardo VM Jr, Anacleto R, Parween S, Samson I, de Guzman K, Alhambra CM, Misra G, Sreenivasulu N (2017) Systems genetics identifies a novel regulatory domain of amylose synthesis. Plant Physiol 173(1): 887–906